

Surface TiO₂ and Lunar Crustal Magnetism

Spatial Co-location Analysis - Results

EDUCATIONAL & AI-ASSISTED PROJECT

This is a personal project created for learning and hands-on experience. Artificial Intelligence (Large Language Models) was used extensively throughout its development to help write the code, understand the geophysical concepts, and rigorously catch statistical and logical errors.

This report analyses whether a present-day surface TiO₂ retrieval spatially co-locates with lunar crustal magnetism. The retrieval is mare-calibrated; the full declared footprint includes out-of-domain highlands, and a separate USGS mare-proxy sensitivity is reported. It does not test the temporal/thermal intermittent-dynamo mechanism of Nichols et al. (2026). The impact-antipode feature is a literature benchmark, not an exhaustive rival. Binary XGBoost/SHAP results are legacy diagnostics. Their scores use 30x30 degree spatially-blocked GroupKFold (seed 42, grid 1.0 deg, age mask 'imbrian', threshold 5.0 nT).

Data mode: real

Run mode: full

Transparent Continuous-Field Analysis

Controls-only blocked R²: 0.2590 +/- 0.1377

Controls + TiO₂ blocked R²: 0.2712 +/- 0.1366

Incremental TiO₂ R²: +0.0121 +/- 0.0052

Fold scores and coefficients are descriptive; no row-level p-value is valid when the effective number of independent spatial regions is small.

Legacy Threshold Classifier (PR-AUC)

Positive prevalence: 0.014 (n=20,556 px, 72 blocks)

Dummy (stratified): 0.0147

Dummy (prior): 0.0139

Logistic regression: 0.0832

XGBoost H₂-only: 0.1106

XGBoost full: 0.0890

XGBoost full (nested-tuned): 0.1009

Legacy Ablation (surface-TiO₂ proxy)

Full: 0.0890 All Ti-derived terms removed: 0.0920

Mean drop from clean TiO₂ ablation: -0.0030

Paired Wilcoxon p (full > no-TiO₂): 0.7812

Exploratory (non-Nichols) gravity/interaction drop: 0.0150 (expected ~0)

Spatial-Rotation Permutation Test

Observed full PR-AUC: 0.0890 Null mean: 0.1127 +/- 0.0569

Empirical p-value: 0.5644 (n=100)

Fold-matched rotations: 100 accepted / 125 candidates; 12 rejected

Exploratory SHAP: TiO₂ vs antipode benchmark

H1 TiO₂ FAMILY |SHAP| (composition, 25/50/100 km combined): 0.4792

H2 antipode feature |SHAP|: 0.9898

Exploratory Ti x gravity family |SHAP| (non-Nichols): 0.3858

H1 TiO₂ signal outranks antipode: False

Legacy Repository-Plan Criteria

(i) beats null + all baselines: False

(ii) TiO₂ signal outranks antipode: False

(iii) TiO₂ ablation drop significant: False

Spatial adequacy gate for a POSITIVE (support survives largest block): False

Spatial Dependence Diagnostics

Full PR-AUC 0.054, conditional block interval [0.013, 0.131] over 72 blocks

Conditional block-resampling interval only; selected blocks are smaller than the estimated spatial range and/or too few effective regions exist.

Large-scale structure range 3752 km; CV block 910 km

Phase-randomised null (strict): $p = 0.686$ ($n=50$)

Partition sensitivity: proxy support at largest CV block = False

CAVEAT: low prevalence / few effective independent regions -- this negative result may be underpowered to detect a weak true effect and absence of a detected effect is not evidence of absence.

Positive-Control Detection Power

Injected signal recovered somewhere on tested grid: True

Observed impact-antipode benchmark recovered: True

Injected + observed benchmark recovery checks passed: True

Power at target effect: not estimable (no justified target effect)

Adequate power demonstrated: False

First tested latent coefficient reaching 80% point power: 1

Conservative 95% floor: 2.0

The injected curve quantifies a design-conditional detection floor. No externally justified target effect was declared, and the observed impact-antipode benchmark must also clear its own spatial null.

Mare-Validity Sensitivity

Mask source: USGS Unified Geologic Map of the Moon 1:5M GIS v2, GeoUnits

Mare rows / positives: 6232 / 58

Mare-only TiO₂ increment (PR-AUC): +0.0676

Positive spatial blocks: 9 paired p : 0.2188

Mare-only continuous TiO₂ incremental R²: +0.0127 +/- 0.0126

USGS 1:5M geologic units are a generalized surface-geology proxy, not the formal pixel-level validity domain published for the LROC WAC calibration. Exact mare-unit polygons conservatively exclude mapped superposed crater, ejecta, plains, and dark-mantling units even where they occur inside a mare. Map-unit boundaries and their rasterization are resolution dependent; edge pixels can be

classification mixtures. The row-local geology proxy restricts this sensitivity to raw TiO₂ only; it does not establish formal WAC validity or support for buffered TiO₂ features.

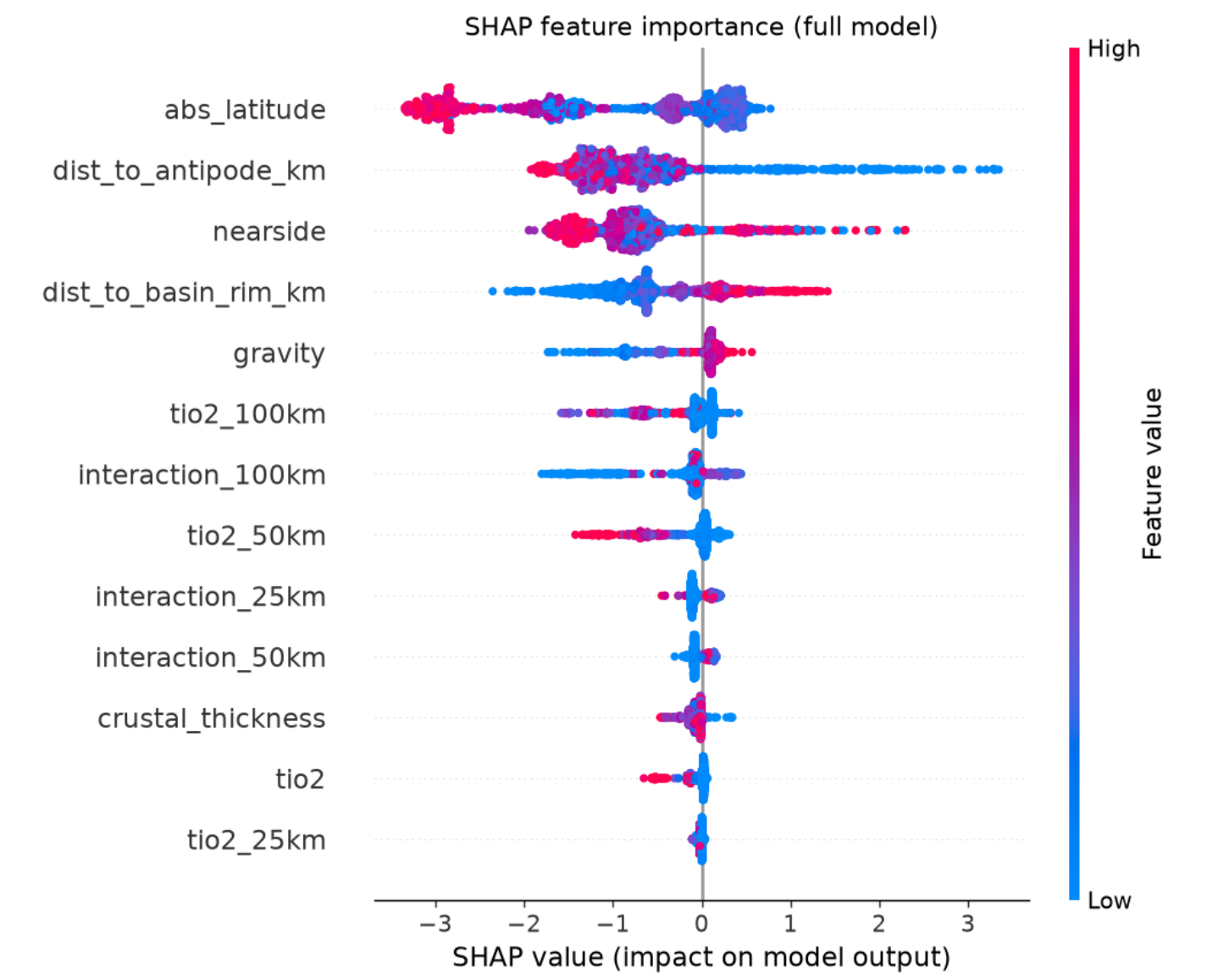
Conclusion: INCONCLUSIVE (LOW DETECTION POWER)

The Nichols et al. temporal intermittent-dynamo mechanism is not adjudicated by this present-day spatial analysis.

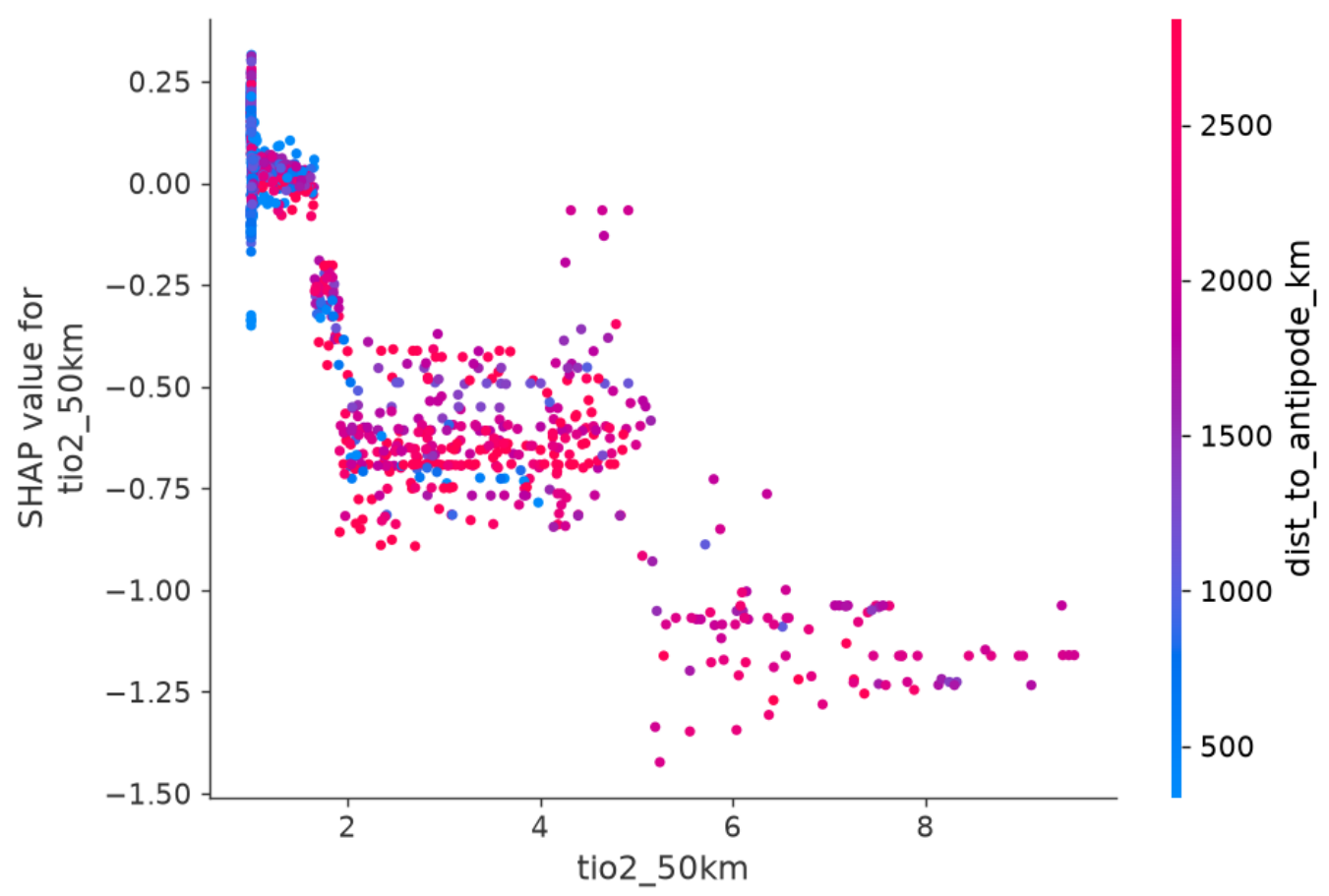
Scope and Caveats (see Fallacy-Audit.md)

- Nichols et al. analyse dated samples and a time-dependent core-mantle mechanism. Present-day surface co-location is an additional proxy proposed by this repository, not a prediction stated by those authors.
- A dynamo is global and magnetises cooling material during active epochs; source age and cooling history are absent from this analysis.
- The sample relationship and pixel-scale regolith map are different estimands; no ecological scaling conclusion follows from this result.
- The remanence carrier is fine-grained metallic iron, not ilmenite; surface TiO₂ is an indirect proxy for the strong-field basalts (F4).
- The TiO₂ proxy and antipode benchmark are not exhaustive hypotheses (long-lived dynamo, impact-plasma amplification, basal-magma-ocean dynamo also exist). This study DISCRIMINATES two signatures; it does not adjudicate the whole field (F5).
- The nearside/Procellarum setting confounds Ti with basin proximity and thin crust; a 'nearside' control plus spatial-block CV mitigate but do not remove it (F8).
- Synthetic injection results measure this pipeline's sensitivity under declared generative assumptions; they are not evidence about the Moon.
- The repository plan has no independent timestamp. Its post-hoc inference amendment is disclosed rather than presented as externally verified pre-registration.

SHAP Summary (feature importance)



SHAP Dependence: TiO2 (H1) coloured by antipode (H2)



Global Map: where the TiO2 (H1) signal drives predictions

